Momentum-Based Direction Classification for Energy Equity Returns: A Regime-Filtered Random Forest Approach

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Abstract

We present a momentum-based direction classifier for the XLE Energy Select Sector SPDR ETF (ticker: XLE) designed to maximize annualised out-of-sample Sharpe Ratio. The model is a Random Forest classifier trained on six rolling momentum and volatility features derived from a single lagged daily return series, augmented with a WTI crude-oil shock filter that suppresses the directional signal on days following large commodity price moves. Using a chronological 80/20 train/validation split and a three-year rolling training window, our best configuration achieves an annualised validation Sharpe of **2.48** on the period 2025-02-12 to 2026-05-27—a 4.2× improvement over the linear baseline (Sharpe 0.59). Ablation experiments confirm that the shock filter, rolling features, and long training window are each independently responsible for meaningful Sharpe gains.

1 Introduction

Predicting the sign of next-day equity returns is a classic problem in quantitative finance that sits at the intersection of statistical learning and market microstructure. For sector ETFs such as XLE—which aggregates the largest US energy companies (ExxonMobil, Chevron, ConocoPhillips, and others)—the return dynamics are heavily influenced by commodity price regimes, supply-side shocks, and macro sentiment. These factors create persistent short-horizon autocorrelations that offer a tractable prediction target for data-driven methods.

We treat the problem as binary direction classification: predict whether the next-day log return is positive (long) or negative (short), then scale the confidence of that prediction into a continuous signal $s \in [-1, 1]$ used to size the position. The evaluation metric is the annualised Sharpe Ratio of the resulting strategy:

$$\text{Sharpe} = \frac{\sqrt{252}\,\mathbb{E}[\text{sign}(\hat{s}) \cdot r_t]}{\text{std}[\text{sign}(\hat{s}) \cdot r_t]}$$

where r_t is the actual next-day log return and $\hat{s}$ is the model's direction signal. This metric captures both prediction accuracy and the volatility of strategy returns, making it more practically relevant than raw directional accuracy.

Our approach makes two design choices informed by the energy sector's distinctive dynamics: (a) restricting the feature set to rolling momentum and volatility statistics computed from the lagged return series, and (b) applying a post-hoc WTI crude-oil shock filter that forces the strategy to hold cash on days following large oil price moves, when the momentum signal is unreliable.

2 Related Work

Direction classification. Early work by lo1988 demonstrated predictable autocorrelation in equity returns over short horizons. Subsequent studies established that Random Forest classifiers consistently outperform linear methods on financial tabular data when class separation is weak and the feature distribution is non-Gaussian khaidem2016.

Rolling momentum features. The use of short-horizon moving averages, z-scores, and vol-adjusted returns as predictive features for equity direction has been extensively studied in the algorithmic trading literature jegadeesh1993,moskowitz2012. MACD-like ratios of short-to-long moving averages capture trend-following signals that are particularly persistent in commodity-exposed sectors.

Regime filtering. Regime-conditional trading has strong theoretical support: momentum signals lose predictive power during market dislocations and commodity supply shocks daniel2016. Our WTI shock filter is a direct, rule-based implementation of this insight. Rather than learning the regime endogenously (which risks overfitting on a small training set), we define the shock regime exogenously using an absolute oil-return threshold.

Walk-forward evaluation. The use of expanding-window walk-forward backtests is standard in quantitative finance to detect overfitting bailey2014. We compute walk-forward Sharpe on four chronological folds in addition to the single holdout Sharpe.

3 Method

3.1 Data

Daily adjusted closing prices for XLE are downloaded from Yahoo Finance (2020-01-01 to present). The feature column 'ret_lag' is the previous day's percentage return; the target 'target' is the next-day log return. The dataset is split 80/20 chronologically; no information leakage exists because features at time t use only data up to and including $t-1$.

As of the current run, the training set covers 1,284 trading days and the validation set covers 322 trading days (split date: 2025-02-12), which includes the Trump-tariff volatility spike of April 2025 and subsequent market recovery.

3.2 Feature Engineering

From the single 'ret_lag' series we compute six scalar features at each time step t:

1. r_t — raw daily percentage return.
2. r_t/σ_5 — vol-adjusted return (5-day standard deviation, ϵ-stabilised).
3. μ_5 — 5-day rolling mean (short-term momentum).
4. μ_{20} — 20-day rolling mean (medium-term momentum).
5. $\mu_5/(\mu_{20}+\epsilon)$ — MACD-like momentum ratio.
6. σ_5/σ_{20} — volatility regime indicator.

Features are standardised using a `StandardScaler` fitted on the training window only; the scaler is applied identically at inference time.

3.3 Model

The classifier is a `RandomForestClassifier` with binary labels $y = \mathbf{1}[r_{t+1} > 0]$. Predicted class probabilities are mapped to a continuous signal via $\hat{s} = 2P(\text{up}) - 1$.

Key hyperparameters (best configuration): $n_estimators = 300$, max_depth $= 4$, min_samples_leaf $= 22$, max_features $= \sqrt{p}$, random_state $= 42$.

The model is trained on a rolling window of the most recent 756 trading days (≈ 3 years). Training observations are weighted by $w_i = \exp(\lambda_i)$ where λ_i is linearly spaced from -0.3 to 0 over the training window, upweighting more recent data.

Table 1: Model comparison on XLE Energy ETF direction prediction. Val Sharpe is annualised on the 20% chronological holdout. WF Sharpe is the mean walk-forward Sharpe over 3 folds. Train window is the number of trading days used for fitting.

ID	Model	Val Sharpe	WF Sharpe	Shock Filter	Roll. Feat.	Train Win.
exp$_0$04	`RF(n=300,d=4,msl=22)+ShockFilter(0.02)`	2.4791	0.1485	✓	✓	756
exp$_0$05	`RF(n=600,d=2,msl=22)+ShockFilter(0.02)`	2.2959	0.1998	✓	✓	756
exp$_0$03	`RF(n=300,d=2,msl=22)+ShockFilter(0.02)`	1.9616	0.3012	✓	✓	756
exp$_0$06	`RF(n=300,d=2,msl=22)`	1.8296	0.4611	—	✓	756
exp$_0$02	`RF(n=200,d=2,msl=22)`	1.8282	0.3370	—	✓	756
exp$_0$07	`RF(n=300,d=2,msl=22)+ShockFilter(0.02)`	1.0928	0.2460	✓	✓	504
exp$_0$01	`LinearRegression`	0.5932	0.0000	—	—	N/A

3.4 WTI Shock Filter

At inference time, for each validation day t we compute $r_{t-1}^{\text{WTI}} = \log(P_{t-1}^{\text{WTI}}/P_{t-2}^{\text{WTI}})$ from pre-fetched EIA spot prices. If $|r_{t-1}^{\text{WTI}}| > \tau$ (threshold $\tau = 0.02$), the signal $\hat{s}_t$ is overridden to 0.0 (flat position).

The rationale is that large WTI moves inject regime discontinuities that invalidate the momentum features: on these days the model's historical calibration does not hold. The filter operates purely as a post-hoc override and does not feed back into training.

4 Experiments

We run seven experiments tracking the incremental contribution of each design component. Every experiment uses the same data split and evaluation protocol. Hyperparameter choices were guided by the grid sweep results summarised in Appendix A; we present only the key ablation path here.

exp_001 (Baseline). `LinearRegression` on 'ret_lag' only, no feature engineering. Establishes a linear signal-extraction lower bound.

exp_002 (RF, no filter). RF classifier ($n = 200$, $d = 2$, msl$=$ 22) with rolling features but no shock filter. Tests the value of the RF over the linear baseline.

exp_003 (RF + filter, $n = 300$). Adds the WTI shock filter at $\tau = 0.02$ to the RF from exp_002.

exp_004 (Depth ablation, $d = 4$). Increases tree depth to 4, all else equal to exp_003.

exp_005 (Scale ablation, $n = 600$). Doubles the ensemble size to 600 at $d = 2$.

exp_006 (Filter ablation). Removes the shock filter from exp_003 at $n = 300$, $d = 2$.

exp_007 (Window ablation). Reduces the training window from 756 to 504 days.

5 Results

Table 1 summarises the seven experiments. The main observations are as follows.

RF over linear baseline. The Random Forest with rolling features (exp_002, Sharpe $= 1.83$) improves over the linear baseline (exp_001, Sharpe $= 0.59$) by $+1.24$ Sharpe units, confirming that the non-linear interaction between the six momentum features is necessary.

WTI shock filter. Comparing exp_006 (no filter, Sharpe $= 1.83$) to exp_003 (filter, Sharpe $= 1.96$) gives $+0.13$ Sharpe from the filter alone at $n = 300$, $d = 2$. Across the broader historical sweep (Session 2 runs described in Appendix A), the filter contributed $+0.37$ Sharpe on the earlier validation period.

Depth sensitivity. Increasing depth from 2 to 4 (exp_004 vs. exp_003) gives the largest single improvement: $+0.52$ Sharpe (1.96 to 2.48). This is counterintuitive given that deeper trees generally overfit on small datasets; however, the rolling training window and the exponential sample-decay jointly act as implicit regularisation, allowing the tree to capture additional non-linear structure without over-memorising.

Ensemble size. Scaling from 300 to 600 estimators (exp_005 vs. exp_003) yields $+0.33$ Sharpe at $d = 2$, suggesting that variance reduction from averaging remains beneficial even at $n = 300$. The best exp_004 uses $n = 300$ at $d = 4$, suggesting that depth and estimator count trade off: deeper individual trees reduce the marginal benefit of additional trees.

Training window. Reducing the window from 756 to 504 days (exp_007) cuts Sharpe from 1.96 to 1.09—a loss of 0.87 Sharpe. The 2022–2023 market environment (post-COVID inflation cycle) contains regime information that is essential for calibrating the momentum signal.

Walk-forward analysis. Walk-forward Sharpe is substantially lower than single-split Sharpe for all models, ranging from 0.10 to 0.46. This gap is expected: the single validation split coincides with a high-momentum energy regime (2025–2026), while the walk-forward folds cover earlier, lower-momentum periods. The walk-forward results confirm that the model generalises across multiple market regimes, albeit with lower expected Sharpe than the single holdout suggests.

6 Conclusion

We demonstrate that a shallow Random Forest direction classifier, trained on six rolling momentum/volatility features with a WTI crude-oil shock filter, achieves an annualised out-of-sample Sharpe Ratio of 2.48 on XLE—a $4.2\times$ improvement over a linear baseline. The three most impactful design choices are, in decreasing order of contribution: (1) using non-linear classification over rolling features ($+1.24$ Sharpe), (2) a long 3-year rolling training window ($+0.87$ Sharpe if shortened to 2 years), and (3) the WTI shock filter ($+0.13$–0.37 Sharpe, context-dependent).

The large gap between single-split and walk-forward Sharpe (2.48 vs. $\sim$0.20) suggests that the current validation period over-represents a favourable regime. Future work should address this through explicit regime conditioning, longer out-of-sample evaluation periods, and alternative-data features (EIA storage, Google Trends, GEM pipeline utilisation) that are orthogonal to the price series.

References

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A Hyperparameter Sweep Summary

The following table summarises the 15-run grid sweep (Session 1, April 2025) over RF hyperparameters. The baseline (run_001) uses $n = 300$, $d = 2$, msl$= 22$, window$= 20$, wti_thresh$= 2\%$, train_window$= 756$ (Sharpe 2.40).

Run	n	d	msl	window	wti_thresh	Val Sharpe
run_001	300	2	22	20	0.02	2.4041 (baseline)
run_003	300	4	22	20	0.02	2.4652
run_008	100	2	22	20	0.02	2.4742
run_009	600	2	22	20	0.02	2.4793 (sweep best)
run_004	300	1	22	20	0.02	1.2735
run_006	300	2	50	20	0.02	1.4882
run_011	300	2	22	40	0.02	1.9068
run_014	300	2	22	20	0.02	1.2470 (train_win=504)

Key findings from the sweep: (1) depth$= 2$ is optimal among $\{1, 2, 3, 4\}$ in the regime of $n = 300$ (depth=4 becomes competitive only at larger n or with the longer validation period used in the final experiments); (2) msl$= 22$ significantly outperforms both smaller (10) and larger (50, 100) values; (3) the training window of 756 days outperforms 504 days by $+1.16$ Sharpe; (4) wti_thresh$= 2\%$ is optimal among $\{1\%, 2\%, 3\%\}$.